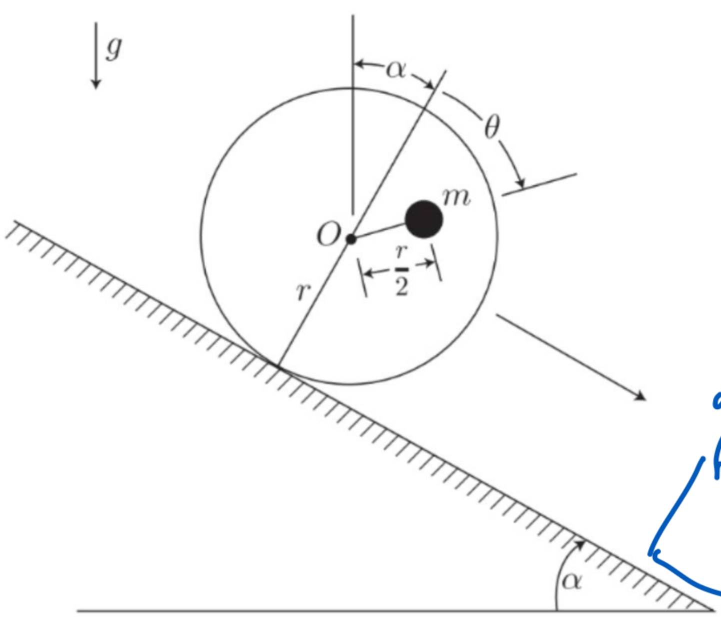
Problem 1

A disk of radius r and negligible mass (i.e., assume the disk is massless) has a particle of mass m embedded at a distance $\frac{1}{2}r$ from the center O, as shown in the figure. Assume the disk rolls without slipping down a plane inclined at an angle α from the horizontal. Use θ as the single degree of freedom for this system. Derive the Hamilton's equations of motion for θ, p_θ.



$x\,x \cdot x\,y$

$x^2 + x\,y + y^2$
$+ y\,x$

$\frac{2}{4} + \frac{1}{4} \cdots$

$$\vec{r}_0 = r\theta\,\hat{n}_1 + r\,\hat{n}_2$$

$$\vec{r}_m = \vec{r}_\theta + \frac{r}{2}\sin\theta\,\hat{n}_1 + \frac{r}{2}\cos\theta\,\hat{n}_2$$

$\hat{n}_2$

$\hat{n}_1$

$$\dot{r}_m = \left(r\dot\theta + \frac{r}{2}\dot\theta\cos\theta\right)\hat{n}_1 - \left(\frac{r}{2}\dot\theta\sin\theta\right)\hat{n}_2$$

$$|\dot{r}_m| = \left(r^2\dot\theta^2 + \frac{r^2}{2}\dot\theta^2\cos\theta\right)^2 + \frac{r^2}{4}\dot\theta^2\cos^2\theta$$

$$|\dot{r}_m| = r^2\dot\theta^2\left(\left(1 + \frac{1}{2}\cos\theta\right)^2 + \frac{1}{4}\sin^2\theta\right)$$

$$1 + \cos\theta + \frac{1}{4}\cos\theta^2 + \frac{1}{4}\sin^2\theta$$

$$|\dot{r}_m| = r^2\left(\frac{5}{4} + \cos\theta\right)\dot\theta^2$$

$$T = \frac{1}{2} m r^2 \left(\frac{5}{4} + \cos\theta\right) \dot{\theta}^2$$

$$I(\theta) = m r^2 \left(\frac{5}{4} + \cos\theta\right)$$

$$T = \frac{1}{2} I(\theta)\, \dot{\theta}^2$$

$$\vec{g} = g\left(\sin\alpha\, \hat{n}_1 + \cos\alpha\, \hat{n}_2\right)$$

$$\vec{r} = \left(r\theta + \frac{r}{2}\sin\theta\right)\hat{n}_1 - \left(r + \frac{r}{2}\cos\theta\right)\hat{n}_2$$

$$V = -m\, \vec{g} \cdot \vec{r}$$

$$mgr\left(\sin\alpha\,\theta + \frac{1}{2}\sin\alpha\sin\theta - \cos\alpha + \frac{1}{2}\cos\alpha\cos\theta\right.$$

$$= mgr\left[\sin\alpha\left(\theta + \frac{1}{2}\sin\theta\right) + \cos\alpha\left(1 + \frac{1}{2}\cos\theta\right)\right]$$

$$L = \frac{1}{2} I(\theta)\dot{\theta}^2 - V(\theta)$$

$$p_\theta = \frac{dL}{d\dot{\theta}} = mr^2\left(\frac{5}{4} + \cos\theta\right)\dot{\theta}$$

$$H = T + V \quad \left(\begin{array}{c} \text{quadratic} \\ \text{Time independent} \end{array}\right)$$

$$\frac{1}{2} I\theta\dot{\theta}^2 + V(\theta)$$

$$=$$

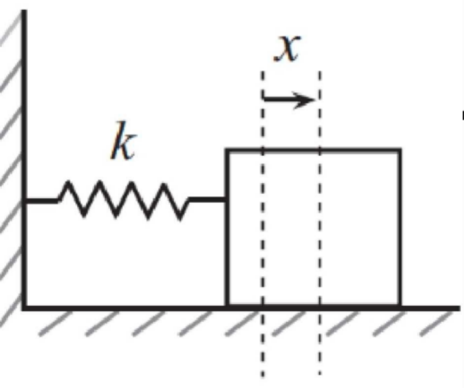
$$\frac{1}{2} \cdot \frac{p_\theta^2}{mr^2\left(\frac{5}{4} + \cos\theta\right)} + mgr\left[\begin{array}{c} -\sin\alpha\left(\theta + \frac{1}{2}\sin\theta\right) \\ + \\ \cos\alpha\left(1 + \frac{1}{2}\cos\theta\right) \end{array}\right]$$

$$\dot{\theta} = \frac{\partial H}{\partial p_\theta} = \frac{p_\theta}{m r^2 \left(\frac{5}{4} + \cos\theta\right)}$$

$$\dot{p}_\theta = -\frac{\partial H}{\partial \theta} = \frac{-p_\theta^2 \sin\theta}{2 m r^2 \left(5/4 + \cos\theta\right)}$$

$$+ m g r \left[-\sin\alpha \left(r + \frac{1}{2}\cos\theta \right) + \frac{1}{2}\cos\alpha \sin\alpha \right]$$

Problem 2



When we use Hamiltonians, we can define canonical coordinate changes (under certain conditions) from (q, p) to (Q, P) by using *generating functions* $F(q, Q, t)$. With this function in terms of the old and new coordinates, the conjugate momenta are defined by,

$$p = \frac{\partial F}{\partial q}, \quad P = -\frac{\partial F}{\partial Q}.$$

Using these relationships, we can write a new Hamiltonian $H'(Q, P)$, which follows the same canonical equations of motion.

$$\dot{Q} = \frac{\partial H'}{\partial P}, \quad \dot{P} = -\frac{\partial H'}{\partial Q}$$

As an example, consider a simple harmonic oscillator with mass $m = 1$ and a spring coefficient k. Let $\omega = \sqrt{k}$. Let $q = x$ be our only generalized coordinate.

a) Derive the Hamiltonian for this system in terms of q, p, ω in the usual way. You should find that,

$$H(q, p) = \frac{1}{2}\left(p^2 + \omega^2 q^2\right).$$

b) Choose the following generating function (I know it seems ridiculous, trust me):

$$F(q, Q) = \frac{1}{2}\omega q^2 cot(2\pi Q)$$

c) Use the relationships above to derive $p(q, Q)$ and $P(q, Q)$.

d) From the expression for P, you solve for $q(Q, P)$, and then substitute this into the expression for p to find $p(Q, P)$.

e) Now find $H'(Q, P) = H(q(Q, P), p(Q, P)$. ($H'$ is a new Hamiltonian in terms of Q and P, not a derivative.)

f) Write the equations of motion of Q, P. You will find that $\dot{Q}$ =constant and $\dot{P} = 0$!

g) Integrate to find $Q(t), P(t)$ and plug into $q(Q, P), p(Q, P)$. You will have found functions of time (in terms of the constant P)! (One textbook author calls this ``cracking a peanut with a sledgehammer.'')

There are three other types of generating functions ($F(q, P), F(p, Q), F(p, P)$) that can be used to canonically change coordinates under certain conditions. We won't fully go into these approaches in this class, but I wanted to show a simple example.

$\vec{r} = x\hat{n}_1$

$\dot{\vec{r}} = \dot{x}\hat{n}_1$

$|\dot{r}| = \dot{x}^2$

$T = \frac{1}{2}m\dot{x}^2 = \frac{\dot{x}^2}{2}$

$V = \frac{1}{2}kx^2 = \frac{1}{2}\omega^2 x^2$

$L = T - V$

$P_x = \frac{\partial L}{\partial \dot{x}} = \dot{x}$

$H = T + V = \frac{P_x^2}{2} + \frac{1}{2}\omega^2 x^2$

a) $= \frac{1}{2}\left(P_x^2 + \omega^2 q^2\right)$

$q = x$

b) $F(q, Q) = \frac{1}{2} \omega q^2 \cot(2\pi Q)$

c) $p(q, Q) = \frac{\partial F}{\partial q} = \omega q \cot(2\pi Q)$

$$P = -\frac{\partial F}{\partial Q} = \frac{1}{2} \omega q^2 \left(-2\pi \csc^2(2\pi Q)\right)$$

$$= \frac{\pi \omega q^2}{\sin^2(2\pi Q)}$$

d) $\therefore \quad q = \sqrt{\frac{P}{\pi \omega}} \sin(2\pi Q)$

$\frac{\cos}{\sin}$

$$\therefore \quad p = \omega \cot(2\pi Q) \sqrt{\frac{P}{\pi \omega}} \sin(2\pi Q)$$

$$= \omega \sqrt{\frac{P}{\pi \omega}} \cos(2\pi Q$$

$$\therefore \quad p = \sqrt{\frac{\omega P}{\pi}} \cos(2\pi Q)$$

e) $H = \frac{1}{2}(p^2 + w^2 q^2)$

cancel

$p^2 = \frac{wP}{\pi}\cos^2(2\pi Q)$

$q^2 = \frac{P}{\pi w}\sin^2(2\pi Q)$

$\therefore H = \frac{wP}{2\pi}$

F) $\dot{Q} = \frac{\partial H}{\partial P} = \frac{w}{2\pi}$ (cons)

$\dot{P} = \frac{\partial H}{\partial Q} = 0$

G) $P(t) = P_0$ $\quad Q = \frac{w}{2\pi}t + Q_0$

$$q = \sqrt{\frac{P}{\pi \omega}} \sin(2\pi Q)$$

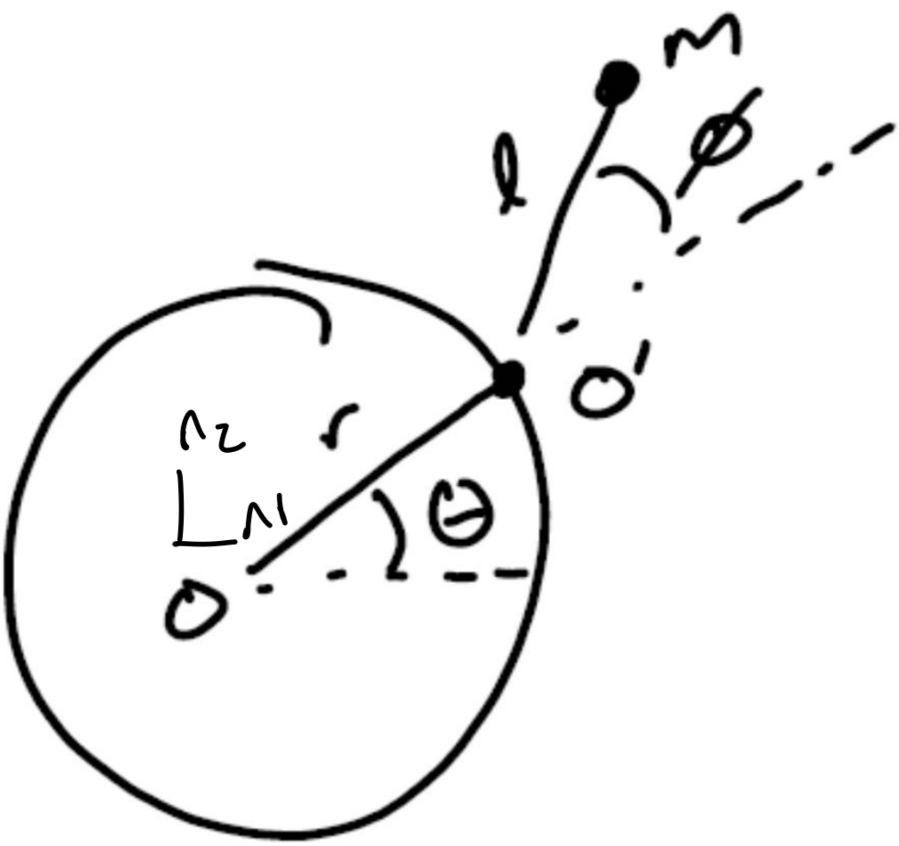
$$q(t) = \sqrt{\frac{P_0}{\pi \omega}} \sin(\omega t + Q_0)$$

$$p(t) = \sqrt{\frac{\omega P_0}{\pi}} \cos(\omega t + Q_0)$$

Problem 3

A particle of mass m is attached by a string of length l to a point O' which moves around a hoop of radius r at a constant angular rate $\dot{\theta} = \omega_0$. The angle between the radial direction and the string is given by ϕ. This setup is moving horizontally on a table top (i.e., gravity points into the page).

Apply Hamilton's Equations of Motion for the generalized coordinate $q = \phi$ and its conjugate momentum p.



$\vec{x} = r\cos\theta + L\cos(\theta+\phi)$

$\dot{\vec{x}} = -r\sin\theta\dot{\theta} - L(\dot{\theta}+\dot{\phi})\sin(\theta+\phi)$

$\dot{\theta} = \omega_0 \therefore$

$\dot{\vec{x}} = -r\sin\theta\,\omega_0 - L(\omega_0+\dot{\phi})\sin(\theta+\phi)$

$$|\dot{\vec{x}}| = r^2\sin^2\theta\omega_0^2$$

$$+2rL(\omega_0+\dot{\phi})\sin\theta\sin(\theta+\phi)$$

$$+L^2(\omega_0+\dot{\phi})^2\sin^2(\theta+\phi$$

Do this for Y, you get the same but with cos.

b/c $v^2 = x^2 + y^2$

$$|\dot{\vec{r}}| = (r\omega_0)^2 + (L(\omega_0+\dot{\phi}))^2$$

$$+2(r\omega_0)(L(\omega_0+\dot{\phi})\cos\phi$$

$$V = -m\vec{g}\cdot\vec{r} = 0$$

$$\therefore$$

$T = L = H =$

$$\frac{1}{2}m\left[(r\omega_0)^2 + \left((L(\omega_0+\dot{\phi})\right)^2 + 2(r\omega_0)(L(\omega_0+\dot{\phi})\cos\phi\right]$$

$\omega_0^2 + 2\omega_0\phi + \phi^2$

$=$

$$\frac{m}{2}r^2\omega_0^2 + \frac{m}{2}L^2(\omega_0+\dot{\phi})^2$$

$$+ m r \omega_0 L \cos\phi(\omega_0 + \dot{\phi})$$

$=$

$$\frac{m}{2}r^2\omega_0^2 + \frac{m}{2}L^2\omega_0^2$$

$$+ mL^2\omega_0\phi + \frac{m}{2}L^2\phi^2$$

$$+ m r L \omega_0 \cos\phi(\omega_0+\dot{\phi})$$

$$\omega_0^2 \frac{m}{2}(\omega_0^2 + L^2)$$

$$+$$

$$\frac{m}{2} L^2 \dot\phi^2 + mL^2 \omega_0 \dot\phi$$

$$+ mrL\omega_0 \cos\phi(\omega_0 + \dot\phi)$$

$$mrL\omega_0^2 \cos\phi + mrL\omega_0 \cos\phi\,\dot\phi$$

$$\therefore$$

$$L = \frac{1}{2} mL^2 \dot\phi^2 + mL\omega_0(1 + r\cos\phi)\dot\phi$$

$$+ \frac{m\omega_0^2}{2}(r^2 + L^2 + 2rL\cos\phi)$$

$$P_\phi = \frac{\partial L}{\partial \dot{\phi}} = mL^2\dot{\phi} + mLw_0(1 + r\cos\phi)$$

$$= mL^2(w_0 + \dot{\phi}) + mrLw_0\cos\phi$$

$$H \neq T + V$$

$$H = p\dot{q} - L$$

$$P_\phi \cdot \dot{\phi} = [mL^2(w_0 + \dot{\phi}) + mrLw_0\cos\phi]\dot{\phi}$$

$$= mL^2\dot{\phi}^2 + mLw_0(1 + r\cos\phi)\dot{\phi}$$

$$- \frac{1}{2}mL^2\dot{\phi}^2 + mLw_0(1 + r\cos\phi)\dot{\phi}$$

$$+ \frac{mw_0^2}{2}(r^2 + L^2 + 2rL\cos\phi)$$

$$= \frac{1}{2}mL^2\dot{\phi}^2 - \frac{mw_0^2}{2}(r^2 + L^2 + 2rL\cos\phi)$$

invert momentum

$$ML^2\dot{\phi} = P_\phi - ML\omega_0(1+r\cos\phi)$$

$$\dot{\phi} = \frac{P_\phi - ML\omega_0(1+r\cos\phi)}{ML^2\dot{\phi}}$$

$$\frac{1}{2}ML^2\dot{\phi}^2 = \frac{1}{2}ML^2 \cdot \left[\frac{P_\phi - ML\omega_0(1-r\cos\phi)}{ML^2\dot{\phi}}\right]^2$$

$$H = \frac{\left[P_\phi - ML\omega_0(1-r\cos\phi)\right]^2}{2ML^2} - \frac{M\omega_0^2}{2}(r^2+L^2+2rL\cos\phi)$$

$$\dot{\phi} = \frac{\partial H}{\partial P} = \frac{P_0 - ML\omega_0(1-r\cos\phi)}{ML^2}$$

$$\dot{P}_\phi = \frac{\partial H}{\partial \phi} = -MrL\omega_0\sin\phi(\omega_0+\dot{\phi})$$

$$\frac{dP}{dt} = ML^2\ddot{\phi} - MrL\omega_0\dot{\phi}\sin\phi$$

$$\therefore \ddot{\phi} = -\frac{r\omega_0^2}{L}\sin\phi$$